Critical Incident Paper

Advantages and disadvantages of Artificial Intelligence development as it pertains to its effect on macroeconomics

Mariia Lukashenko

ECO-416

Dr. Baranes

March 19, 2026

Spending on artificial intelligence has grown at a speed that makes it difficult to treat AI as just another business trend or another round of digital innovation - by 2024, corporate AI investment had reached $252.3 billion, private AI investment had risen by 44.5 percent, and the broader surge in AI adoption had become visible across firms rather than only inside the technology sector (Stanford HAI, 2025). In the same period, reported organizational use of AI rose sharply, with 78 percent of surveyed organizations saying they used AI in at least one business function in 2024, up from 55 percent in 2023 (Stanford HAI, 2025). Numbers at that scale matter not only because they show implementation and anticipation of a new technology, but because they suggest that a large amount of capital is now being redirected toward AI models, cloud systems, semiconductors, data infrastructure, and organizational restructuring - which means the AI boom is no longer only a story about firms experimenting with new tools, but a broader macroeconomic development.

That shift also matters because AI is no longer sitting at the edge of business activity. Firms are increasingly using it in planning, coding, logistics, customer support, forecasting, and internal decision-making, and some are already redesigning workflows around it rather than simply layering it onto old systems (McKinsey & Company, 2025). This is important from a macroeconomic point of view because investment in AI is not just spending on software - it is also part of a wider process of capital deepening, in which firms invest in the tools, systems, and organizational changes needed to alter production itself. In that sense, AI spending begins to look less like the purchase of one more digital service and more like the early stage of a wider technological transition across the economy.

From that perspective, the main macroeconomic argument in favor of AI development is fairly clear. If AI functions as a general-purpose technology - something closer to electricity, semiconductors, or the internet than to a narrow software upgrade - then its effects should not remain inside one sector, but should spread across industries and gradually raise productivity, efficiency, and potential output. This is where macroeconomic theory becomes useful: in Solow-style terms, AI can raise the level of output by improving technology and encouraging new capital formation, even if it does not immediately or automatically increase the economy's permanent growth rate. The strongest case for AI, then, is not simply that firms can do certain tasks faster, but that sustained investment in AI may support broader productivity growth across the economy over time (Stanford HAI, 2025).

There are also concrete signs that AI is already affecting the economy beyond the firm level. One example is the sharp increase in demand for advanced chips and compute infrastructure, which has concentrated gains in the upstream parts of the value chain that supply AI development. NVIDIA, one of the clearest beneficiaries of this trend, reported $130.5 billion in fiscal 2025 revenue, up 114 percent from the previous year, with data-center revenue also growing extremely quickly (NVIDIA, 2025). That does not simply show that one company performed well - it shows that AI investment is reshaping demand across semiconductors, computing capacity, and related capital markets. In macroeconomic terms, that can be seen as evidence that AI is already redirecting investment toward sectors tied to digital infrastructure and high-value production.

At the same time, the economic case against AI development is not difficult to see either, and this is exactly where the macroeconomic tension begins. The gains from AI are unlikely to be evenly distributed, especially when the firms adopting it fastest are often the firms that already have scale, capital, access to data, and access to advanced computing (McKinsey & Company, 2025). If that pattern continues, AI may raise total output while still widening the gap between leading firms and lagging firms, and between workers whose skills are complemented by AI and workers whose tasks are easier to automate or reorganize. In other words, AI may improve efficiency and still produce a less equal distribution of national income - which makes it a macroeconomic issue, not just a business strategy issue.

A second concern is that AI investment is beginning to affect physical and energy infrastructure as well as digital markets. The International Energy Agency projects that global electricity consumption by data centers will more than double to around 945 terawatt-hours by 2030, with AI identified as the most important driver of that increase (International Energy Agency, 2025). This matters because it complicates the simple idea that AI will only reduce costs. In sectors like customer service, logistics, and back-office administration, AI may lower production costs or improve efficiency, but at the same time it may increase demand for electricity, cooling capacity, data-center construction, and other infrastructure inputs. So, the same technology that may support productivity growth in the long run may also create new cost pressures, bottlenecks, or regional imbalances in the short run.

There is also the problem of timing. Investment has increased very quickly, but the aggregate payoff is still less visible than the spending itself - in practice, the effects are easier to see in functions like coding, customer service, and routine administrative work than in aggregate productivity growth - which is consistent with the broader historical pattern in which major technologies often require complementary investments before their gains appear clearly in productivity statistics. That delay matters because it creates the possibility that expectations of future returns may rise faster than measured macroeconomic results. If firms and investors continue to commit capital on the assumption of large future gains before those gains are broadly realized, then AI development may support long-run growth potential while also creating short-run instability - including overinvestment, uneven diffusion, labor-market disruption, and a more concentrated pattern of economic gains (Stanford HAI, 2025).

This is what makes AI development a strong macroeconomic critical incident. On one side, AI may support capital deepening, productivity growth, and higher potential output across a wide range of sectors, especially in business functions such as marketing and sales, product and service development, service operations, and software engineering, where AI use is already most common (McKinsey & Company, 2025). From the perspective of the Solow growth model, AI can be seen as a technological improvement that raises the level of output by making labor and capital more productive, even if its effect on the long-run growth rate is less certain. Furthermore, broader growth theory suggests that if AI continues to generate innovation, knowledge spillovers, and ongoing productivity gains, its long-run effects may reach beyond a one-time increase in output. On the other side, AI may intensify inequality, strengthen concentration in the most advantaged firms and upstream suppliers, raise pressure on infrastructure and energy demand, and encourage investment to move faster than realized macroeconomic returns. The central issue, thus, is not whether AI is economically important - the scale of spending and adoption has already answered that question - but whether its effects will become broad, productivity-enhancing, and sustainable across the economy, or whether they will remain uneven, disruptive, and potentially destabilizing. This critical incident therefore focuses on that tension by asking how AI development affects economic growth, national income, labor-market adjustment, inflationary pressure, and the broader structure of the macroeconomy.

Critical Incident Overview

This critical incident examines artificial intelligence as a macroeconomic development rather than only a technological or firm-level change. Its central concern is that AI may support productivity growth, capital deepening, and higher potential output over time, while also creating uneven adjustment across firms, sectors, and labor markets. As a result, AI development may strengthen long-run growth and efficiency, but it may also increase inequality, concentration, and short-run instability if the gains from adoption remain uneven or delayed. The incident therefore focuses on whether AI will function as a broad source of macroeconomic improvement or as a more uneven and disruptive force in the wider economy.

Learning Objectives

By the end of this critical incident, readers should be able to:

1. Explain how a positive AI-driven technology shock affects output, labor demand, wages, and capital-market adjustment using Solow model.
2. Analyze how AI may redistribute value creation and national income across firms, sectors, and workers by applying the smile curve to changes in production and labor-market structure.
3. Use the Solow growth model to evaluate the long-run macroeconomic effects of AI investment, especially the difference between a higher level of output and a permanently higher rate of economic growth.

Questions:

- Question 1

How does a positive AI-driven technology shock affect productivity, labor demand, and capital markets, and why do these changes shape both short-run macroeconomic outcomes and long-run growth?

Define:

A positive technology shock is an improvement in the way the economy transforms inputs such as labor and capital into output. In macroeconomics, this means that firms can produce more with the same resources or produce the same amount with fewer resources. If AI improves forecasting, coding, logistics, data processing, and decision-making, then it can raise productivity across multiple sectors rather than only within one firm. This matters because a technology shock does not stay confined to one part of the economy - it changes the marginal product of labor, affects firms' willingness to hire, influences wages, alters expected returns to investment, and can also affect savings and interest rates through capital-market adjustment.

Diagram:

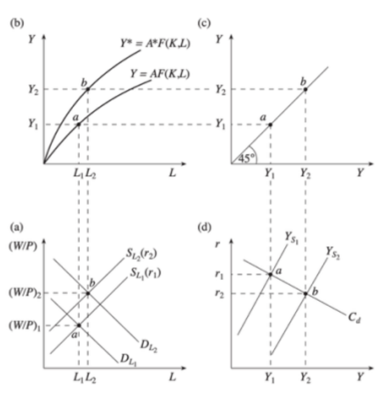


Figure 6.6 The impact of a technology shock

Discuss:

This diagram helps explain how an AI-driven technology shock moves through the economy step by step rather than affecting only one area at a time. In part (b), we can see the production function shift upward from Y = AF (K, L) to Y* = A*F (K, L). This is the clearest starting point for understanding AI. If AI improves the efficiency of production - for example, by making planning, forecasting, coding, logistics, or information processing more effective - then firms can produce more output with the same labor input. In the diagram, that means output rises from Y1 to Y2 as labor increases from L1 to L2. This is directly related to AI because the basic macroeconomic claim behind AI investment is exactly this: that it can shift productivity upward and raise the economy's productive capacity.

That increase in productivity then feeds into the labor market. In part (a), we can see labor demand shift outward from DL1 to DL2. The reason is that once workers become more productive, the marginal product of labor rises, so firms are willing to hire more labor and to pay a higher real wage. The equilibrium moves from point a to point b, which shows both higher employment and a higher real wage. This matters for AI because a positive AI shock does not only change what firms can produce - it also changes how valuable labor becomes inside production. In the short run, that can support higher wages and stronger labor demand in sectors where AI complements workers. At the same time, this effect is unlikely to be uniform across the whole economy, because AI complements some types of labor more than others. So even if the aggregate diagram shows rising labor demand, in practice the gains may be concentrated among workers whose tasks are enhanced by AI, while other workers face displacement or restructuring.

The broader macroeconomic effect becomes clearer in part (c), where the economy moves outward along the 45-degree line from Y1 to Y2. This shows that higher output also means higher income. That connection is important because it shows why AI matters beyond firm-level efficiency. If AI raises output across enough sectors, then it raises total income in the economy, not just the profits of individual firms. In macroeconomic terms, this is one of the main arguments for AI development - that a sufficiently large productivity improvement can raise national output and help support stronger overall economic performance.

In part (d), we can see that the effects of the technology shock also extend into capital markets. The savings schedule shifts from S1 to S2, equilibrium income rises, and the real interest rate changes as the economy adjusts to higher productivity. This is directly connected to AI because AI investment does not involve only software adoption - it also involves spending on semiconductors, cloud systems, data centers, training, and organizational restructuring. As firms expect higher returns from these investments, capital-market conditions begin to adjust as well. In the short run, this can stimulate investment and reinforce expansion. But it can also create tension if expectations about future AI returns rise faster than realized gains, which means that the same technology shock that supports growth can also contribute to overinvestment or imbalance.

Taken together, the diagram shows why AI can shape both short-run macroeconomic outcomes and long-run growth dynamics. In the short run, a positive AI-driven technology shock can raise output, increase labor demand, alter wages, and encourage greater investment through capital-market adjustment. These are the immediate macroeconomic effects of higher productivity. Over the longer run, however, the significance of the shock depends on whether the gains spread broadly across sectors and whether firms make the complementary investments needed to sustain those gains. If AI remains concentrated in only a small set of firms or industries, then the macroeconomic effect will be more limited. But if it diffuses widely enough, then the upward shift in productivity shown in the diagram can support a higher path of output and stronger long-run economic performance.

So, the value of this diagram is that it shows AI not as a narrow innovation, but as a chain reaction across the economy. A positive AI-driven technology shock begins with higher productivity, but it does not stop there - it changes labor demand, income, savings, investment, and the broader trajectory of macroeconomic growth. That is why AI is best understood as a macroeconomic development rather than only a technological one.

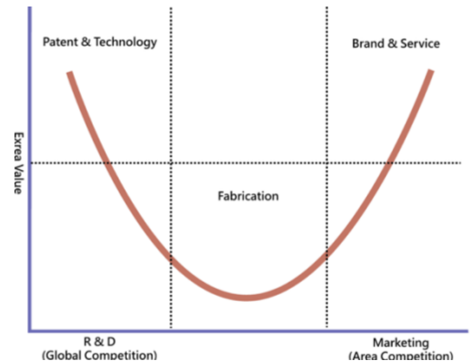
- Question 2

Why does the smile curve help explain how AI may shift value creation and national income toward high-value activities such as R&D, software, and platform-based services, while placing pressure on routine production and middle-skill labor?

Define:

The smile curve is a framework used to show how value is distributed across different stages of production. Its main idea is that value tends to be highest at the early and late stages of the production chain - such as research, design, patents, software, branding, and services - while value tends to be lower in the middle, where fabrication and routine production take place. This concept is useful for AI because AI does not increase value evenly across all stages of production. Instead, it tends to strengthen activities connected to knowledge, data, intellectual property, digital platforms, and customer-facing services, while putting greater pressure on more standardized and routine stages of production. As a result, AI can reshape not only output, but also where profits, wages, and economic power are concentrated.

Diagram:



Discuss:

The smile curve helps explain the macroeconomic effects of AI because it shows that not all parts of production create the same amount of value, and AI tends to reinforce that uneven pattern rather than flatten it. On the left side of the diagram, we can see the part of the curve associated with R&D, patent and technology development. This side matters because AI strongly complements activities that depend on innovation, data systems, model development, software design, and intellectual property. In practical terms, this means that firms operating in these areas are often in a stronger position to capture the gains from AI, since they are the ones creating the models, owning the patents, controlling the software, or developing the systems that other firms then use. This is directly connected to AI because much of the current AI boom is concentrated in exactly these high-value upstream activities.

In the middle of the diagram, the curve falls downward around fabrication. This is the low point of the smile curve, and it represents the more routine, standardized, and cost-pressured stage of production. This part is especially important for understanding the downside of AI. If AI makes it easier to optimize workflows, automate repetitive tasks, standardize processes, and lower the labor needed in routine production, then the middle stage of the curve comes under greater pressure. That does not mean routine production becomes unimportant, but it does mean that it may capture a smaller share of total value. In labor-market terms, this is where middle-skill work often becomes more vulnerable, especially when tasks are easy to codify, monitor, or restructure through AI systems.

On the right side of the diagram, the curve rises again toward branding, marketing, and services. This side of the curve is also highly relevant to AI because AI can strengthen activities connected to customer targeting, platform control, service personalization, advertising, and post-production relationships with consumers. These are all areas where firms can differentiate themselves, strengthen market power, and capture more value even after the physical good has already been produced. In that sense, AI does not only help create things more efficiently - it also helps firms extract more value from distribution, branding, and service-based relationships. This is why the right side of the smile curve becomes more valuable in an AI-driven economy.

Taken together, the diagram shows why AI may redistribute value creation and national income toward the two ends of the production chain while putting pressure on the middle. On the one hand, upstream innovation activities benefit because AI raises the importance of software, patents, algorithms, research, and intellectual property. On the other hand, downstream branding and service activities benefit because AI improves personalization, platform coordination, market access, and consumer data use. But the middle - where production is more routine and more exposed to standardization - may face greater pressure from automation and cost competition.

This matters for macroeconomics because national income is not only determined by how much output the economy produces, but also by who captures the gains from that output. If AI strengthens the high-value ends of the curve while weakening the middle, then profits and wage gains may become more concentrated in firms and workers linked to innovation, platforms, and services, while middle-stage producers and middle-skill workers face greater pressure. In that case, AI may raise overall output and efficiency while also increasing inequality across firms, sectors, and labor groups.

So, the smile curve is useful because it makes the distributional side of AI much clearer. AI is not likely to raise value evenly across the economy. Instead, it may deepen the difference between high-value and routine activities, strengthen the position of firms that control technology and customer access, and shift a larger share of national income toward the parts of the production chain where value capture is already strongest. That is exactly why the smile curve helps explain the macroeconomic tension built into AI development.

- Question 3

How can the Solow growth model be used to explain the long-run macroeconomic effects of AI investment, and why might AI raise the level of output more clearly than the long-run rate of economic growth?

Define:

The Solow growth model explains long-run economic growth by focusing on capital accumulation, labor, savings, depreciation, and technology. In the model, output per worker depends on capital per worker and the level of technology available in the economy. Growth can occur through capital deepening and, more importantly, through technological progress. This model is especially useful for AI because it helps distinguish between two different outcomes. One is an increase in the level of output, which means the economy becomes more productive and can produce more per worker. The other is an increase in the long-run growth rate, which means the economy continues growing faster on a permanent basis. The Solow model suggests that a technology improvement can clearly raise the level of output, but it does not necessarily imply a permanently higher growth rate unless technological progress continues over time.

Diagram:

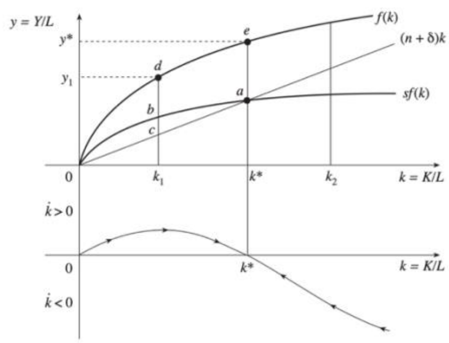


Figure 11.4 The Solow growth model

Discuss:

The Solow diagram helps explain the long-run effects of AI investment by showing how a technology improvement can raise output even if it does not permanently change the economy's growth rate. In the broader Solow framework, output can be written as Y = F(K,AL), where A represents the level of technology. In the diagram, the production function f(k) represents output per worker at each level of capital per worker, and the savings curve sf(k) shows how much of that output is invested. In per-worker terms, this can be written as y = Af (k), which makes clear that an increase in technology raises output for any given level of capital per worker. The line (n+δ)k shows how much investment is needed to keep capital per worker constant, given population growth and depreciation.

At the original steady state, the economy is at k∗, where the savings curve and the break-even investment line intersect. At that point, capital per worker is stable, which means the economy is no longer growing through capital deepening alone. This is the baseline from which we can think about AI. If AI improves productivity, then for any given amount of capital per worker, the economy can produce more output per worker. If Al raises the technology factor from A to A', where A' > A, then output per worker rises from y = Af (k) to y' = A'f (k) at the same level of k. In the diagram, that means the production function shifts upward. As a result, output rises from lower levels such as y1 toward higher levels such as y∗, even if the capital-labor ratio does not immediately change very much. This is directly related to AI because the core macroeconomic argument for AI is that it raises efficiency - firms can generate more output from the same underlying inputs.

The diagram also helps explain why AI can support capital deepening over time. When productivity rises, output rises, and that creates the possibility of higher savings and investment. In the figure, we can see points such as b, d, and e above the lower output levels, which help show that as the economy moves upward, it can sustain a higher level of output per worker than before. This is important because AI investment is not limited to software alone. It also includes semiconductors, cloud systems, data infrastructure, and organizational redesign. These investments can increase the productivity of both labor and capital, helping the economy move toward a higher-output path.

At the same time, the Solow model also shows the limits of that process. Even if AI raises output and encourages additional investment, the economy still faces diminishing returns to capital. That is why the diagram matters so much. As the economy moves toward a new steady state, growth generated by capital deepening eventually slows. In other words, AI can raise the level of output by shifting the economy upward, but once that adjustment is complete, the economy does not automatically continue growing faster forever. This is exactly why AI may raise the level of output more clearly than the long-run rate of economic growth.

That distinction becomes even clearer if we think about the difference between a one-time productivity improvement and ongoing technological progress. If AI acts mainly as a large upward shift in efficiency, then the economy becomes richer - output per worker rises, and the steady-state level is higher. But if AI is to raise the long-run growth rate itself, then technological progress must continue pushing the economy forward on an ongoing basis rather than only once. The Solow framework therefore reminds us that a very large investment boom does not automatically imply permanently faster growth. It may instead imply a very meaningful level effect - a higher path of output, a richer economy, and a more productive capital stock - without changing the long-run growth rate in a permanent way.

This is why the Solow growth model is so useful for understanding AI investment. It allows us to say two things at the same time without contradiction. First, AI can be macroeconomically transformative, because it can raise productivity, support capital deepening, and move the economy to a higher level of output per worker. Second, that does not necessarily mean that AI will permanently accelerate the rate at which the economy grows year after year. The diagram makes that distinction visible: AI shifts the economy upward, but the long-run growth effect depends on whether technological progress continues broadly and persistently enough to keep pushing the production function higher over time.

Epilogue

AI increasingly restructures organizations rather than simply adding another tool to existing workflows. As firms integrate it more deeply, they redesign production processes, shift decision-making, and reorganize how labor, data, and capital are used across the business. That is what makes AI closer to a general-purpose technology — its effects spread across functions and sectors rather than remaining limited to one task or department. At the same time, this restructuring remains uneven. Firms with greater scale, stronger data access, and more computing capacity are better positioned to capture the gains, while slower-moving firms and more routine forms of work face greater pressure. The future significance of this incident therefore depends on diffusion: if these organizational gains spread broadly, AI may support wider productivity growth, but if they remain concentrated, the result is more likely to be an uneven and unequal restructuring of the economy.

Sources:

Acemoglu, D. (2025). *The simple macroeconomics of AI. Economic Policy.*

https://doi.org/10.1093/epolic/eiae042

Alderucci, D., Bloom, N., Falk, B., Fei, Z., Graf, M., Hunter, D., Kohli, R., Osman, M., & Van Reenen, J. (2025). *Macroeconomic productivity gains from artificial intelligence in G7 economies*. OECD Publishing. https://doi.org/10.1787/a5319ab5-en

Bonney, K., Breaux, C., Buffington, C., Dinlersoz, E., Foster, L., Goldschlag, N., Haltiwanger, J., Kroff, Z., & Savage, K. (2024, March). *Tracking firm use of AI in real time: A snapshot from the Business Trends and Outlook Survey*. U.S. Census Bureau, Center for Economic Studies. https://www.census.gov/hfp/btos/downloads/CES-WP-24-16.pdf

Bonfiglioli, A., Crinò, R., Gancia, G., & Papadakis, I. (2025). Artificial intelligence and jobs: Evidence from US commuting zones. *Economic Policy,* https://doi.org/10.1093/epolic/eiae059

Cazzaniga, M., Jaumotte, F., Li, L., Melina, G., Panton, A. J., Pizzinelli, C., Rockall, E., Tavares, M. M., & Yang, J. (2024). *Gen-AI: Artificial intelligence and the future of work*. International Monetary Fund. https://www.imf.org/en/publications/staff-discussion-notes/issues/2024/01/14/gen-ai-artificial-intelligence-and-the-future-of-work-542379

Drago, C., Costantiello, A., Savorgnan, M., & Leogrande, A. (2025). Macroeconomic and labor market drivers of AI adoption in Europe: A machine learning and panel data approach. *Economies*, Article 226. https://doi.org/10.3390/economies13080226

Filippucci, F., Gal, P., Millot, V., & Sorbe, S. (2024). *Miracle or myth? Assessing the macroeconomic productivity gains from artificial intelligence*. OECD Publishing. https://www.oecd.org/en/publications/miracle-or-myth-assessing-the-macroeconomic-productivity-gains-from-artificial-intelligence_b524a072-en.html

International Energy Agency. (2025). *Energy demand from AI*. In *Energy and AI*. https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai

Lane, M., Williams, M., & van der Broecke, S. (2023). *The impact of artificial intelligence on productivity, distribution and growth*. OECD Publishing. https://www.oecd.org/en/publications/the-impact-of-artificial-intelligence-on-productivity-distribution-and-growth_8d900037-en.html

McKinsey & Company. (2025, March 12). *The state of AI: How organizations are rewiring to capture value*. https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai-how-organizations-are-rewiring-to-capture-value

NVIDIA. (2025, February 26). *NVIDIA announces financial results for fourth quarter and fiscal 2025*. NVIDIA Newsroom. https://nvidianews.nvidia.com/news/nvidia-announces-financial-results-for-fourth-quarter-and-fiscal-2025

Pizzinelli, C., Panton, A. J., Li, L., Tavares, M. M., Cazzaniga, M., Rockall, E., Jaumotte, F., & Melina, G. (2024). *Broadening the gains from generative AI: The role of fiscal policies*. International Monetary Fund. https://www.imf.org/en/publications/staff-discussion-notes/issues/2024/06/11/broadening-the-gains-from-generative-ai-the-role-of-fiscal-policies-549639

Stanford Institute for Human-Centered Artificial Intelligence. (2025). *Economy*. In *The 2025 AI Index report*. Stanford University. https://hai.stanford.edu/ai-index/2025-ai-index-report/economy

Snowdon, B., & Vane, H. R. (2005). *Modern macroeconomics: Its origins, development and current state*. Edward Elgar.

Thompson, B. (2014, October 28). *Publishers and the smiling curve*. Stratechery. https://stratechery.com/2014/publishers-smiling-curve/